Is Lotka–Volterra-type of model right for five-element network in TMC?

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Abstract

1 Introduction

Traditional Chinese medicine (TCM) has been widely recognized and becomes an invaluable component in alternative medicines in the west [**?**] though it is not always viewed correctly. Chinese medicine is the result of science over long time as it is the result of continuous critical thinking and extensive clinical observations and testing [**?**]. It is a coherent and independent system of thoughts and practice that have been developed over 2000 years. As Manfred Porkert [**?**] rightly pointed out, Chinese medicine is not less logical than the Western, just less analytical.

TCM is an attractive alternative for several reasons. First, TCM takes holistic approach, which takes into account of not only interactions among different components within the whole body, but also external environment such as season, time of the day, the place of living and so on. This approach is very different from that of modern medicine, which essentially treat a patient as an independent sick organ with a bunch of testing numbers. It is so often that disease spread from one organ to another. TCM has the principle that strengthen the still healthy organ that is directly impact by the ill organ before treat the sick one so to prevent spread. This can only be achieved by knowing the overall relations between the organs. Second, because of holistic approach, TCM natually uses multi-target approach instead of single-target as modern medicine usually does (add citation). This approach could provide a balanced and effective treatment with less side effect. Third, TCM focuses on health, not disease. For example, it is so common that we feel something is going wrong, for example feeling lack of energy, but all medical tests show nothing wrong. TCM focuses on identifying the root problems. As said in Huang Di Nei JIng, the best doctor is the one who bring patient back to true healthy before the patient even gets sick. Finally, TCM is very accessible to average people. In the past, Chinese has some general idea on diagonizing and even able to construct some treatments by themselves for minor problems.

Imagine that if everybody can identify themselves health issue before gets sick, then the immense cost in medical can be saved, the pollution related to drugs can be dramatically reduced, and people can live healthier and longer and happier.

TCM is developed based on theories and principles described in the ancient work Huang Di Nei Jing. The main theories are based on two networks: a five-node network called five-element system, and a two-node network called yin-yang relation. Diagonsis and treatments are conducted based on the dynamics of the two networks. In this work, we aims to introduce and understand the theories from network dynamics point of view.

Network motif was proposed to understand dynamics of complex network systems by looking at small sub-networks that occur more often than usual or functionally building blocks [**?**, **?**]. It is known that the dynamics of a sub-network in general may not reflect the dynamics of the whole system. However, if the sub-network is relatively independent, then its dynamics could be manifested in the dynamics of the whole system. For example, the sub-network is an isolated network, or it does not have input from remaining networks. Interestingly, in TCM, major principles in diagnosis and treatments are based on the sub-networks of the five-element network.

The five-element network have been proposed to be modeled by a Boolean network system [**?**], or a system of differential equations [**?**, 1, **?**, **?**]. We are interested in Lotka–Volterra-type of model that was proposed in [1]. Lin *et al.*[1] studied the dynamics of the network at several special sets of parameters and explore possibility of chaotic dynamics. Their work was carried out from math interest instead of the applications in TMC. To make a model meaningful, we need to make sure the model captures some key characteristics that are observed in applications. For example, a healthy system needs to be stable in the sense that perturb initial conditions, or perturb parameter values, the overall dynamics stay the same.

In this work, we study equilibrium states and their stability of . explore the overall stability as we assume that a healthy human being should have stable dynamics. We thus explore overall stability of the system Lin's model and a modified version of the model.

2 The five-element network and principles in practice

OUTLINE:

1. FIIVE - ELEMENT NETWORK
2. THE PRINCIPLES IN PRACTICE FOR TREATING ILLNESS

The five-element network consists of the five elements: wood, fire, earth, metal and water. The two types

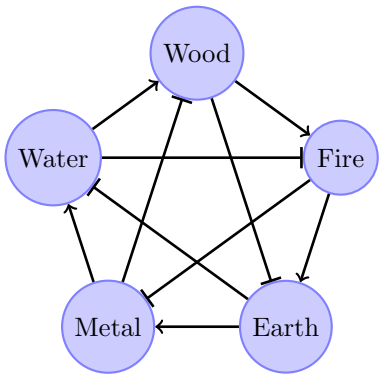


Figure 1: Five-element network and motifs approach in practice

of arrows in the fig. 1 represent two primary inter-element relationships: **Sheng (Generating $\rightarrow$) cycle.** and **Ke (Controlling, $\dashv$) cycle.**

Principles in practice. ADD THOSE DESCRIPTIONS

3 Dynamics of Lin's Model

The numerical results in this section and Section 4 are preliminary comparisons of two complete five-node models under specified parameter choices. They describe the existence and local stability of a positive synchronous equilibrium and responses to selected initial perturbations. Because the models differ in both interaction laws and the presence of a direct child-loss term, these comparisons do not isolate the mechanism responsible for a difference in stability. The coefficients used below are illustrative and have not been calibrated to physiological data.

3.1 Model, balanced equilibrium, and numerical protocol

Let $x_i \geq 0$ denote the activity of node i, with the nodes ordered as Wood, Fire, Earth, Metal, and Water. All indices are taken modulo five. In this ordering, $i-1$ is the generating predecessor (mother), $i+1$ is the generating successor (child), $i-2$ is the controller, and $i+2$ is the controlled target. The homogeneous six-term model of Lin and Han [1] is

$$\dot{x}_i = ax_i - bx_i^2 + cx_ix_{i-1} - dx_ix_{i-2} - gx_ix_{i+1} - hx_ix_{i+2}. \tag{1}$$

Both models studied here have five state variables; "six-term" refers to the contributions to each equation, not to the number of nodes. The nonnegative orthant is invariant for (1) for as long as the solution exists, since every component of its vector field contains x_i as a factor. This property does not imply boundedness or convergence.

We call $x = s\mathbf{1}$ a balanced, or synchronous, state. For $a > 0$, a positive balanced equilibrium exists precisely when $M = b + d + g + h - c > 0$, in which case

$$s_{\mathrm{L}} = \frac{a}{b + d + g + h - c}. \tag{2}$$

At this equilibrium the Jacobian is circulant. With $\theta_j = 2\pi j/5$, its eigenvalue real parts are

$$\operatorname{Re}\lambda_j = s_{\mathrm{L}}\big[-b + (c - g)\cos\theta_j - (d + h)\cos(2\theta_j)\big], \qquad j = 0, \ldots, 4. \tag{3}$$

We use the spectral abscissa $\alpha = \max_j \operatorname{Re}\lambda_j$ to assess local stability. Strict negativity implies local exponential stability, whereas a positive value implies instability. A zero real part requires further analysis. The maps classify $\alpha < -10^{-9}$ as stable, $\alpha > 10^{-9}$ as unstable, and values between these thresholds as unresolved. Absence of a positive synchronous equilibrium is recorded separately; it is not a classification of the system's other equilibria or attractors.

For the trajectory comparisons, we used SciPy's DOP853 integrator with relative tolerance 10^{-10} and absolute tolerance 10^{-12}, and sampled 3,001 times over $0 \le t \le 120$. A repeat with RK45 at the same tolerances changed the sampled activities by at most 1.1×10^{-8} across the four runs. Analytic Jacobians were checked by complex-step differentiation at 1,000 random states per model; the largest absolute entrywise discrepancy was below 8×10^{-15}. These checks support the implementation, but do not establish global dynamical conclusions.

3.2 Baseline stability and response to initial perturbations

At the illustrative Lin baseline $(a, b, c, d, g, h) = (1, 1, 1, 1, 1, 1)$, the positive balanced level is $s_{\mathrm{L}} = 1/3$. The leading pair is approximately $0.206011 \pm 0.634038\,\mathrm{i}$, so this equilibrium is unstable (Figure 2). For comparison, the modified-model spectrum shown in the same figure uses the separate baseline specified in Section 4.2.

We then initialized each model at its own balanced equilibrium and changed only Wood: either $x_1(0) = 1.05s$ or $x_1(0) = 0.30s$, keeping $x_i(0) = s$ for $i = 2, \ldots, 5$. In the Lin runs, the solutions moved away from balance and exhibited sustained oscillatory behavior over the observation interval (Figure 3). The distances $\|x(120) - s_{\mathrm{L}}\mathbf{1}\|_2$ were approximately 0.8776 and 0.9899, respectively. These trajectories illustrate the baseline equilibrium's instability. They do not imply that the model lacks stable equilibria at other parameters, or that an unstable equilibrium necessarily produces unbounded activity or chaos.

3.3 Stability under a two-parameter sweep

To explore parameter dependence, we varied nourishment c and received restraint d on a 90×90 equally spaced grid,

$$c \in [0.3, 1.6], \qquad d \in [0.1, 1.0], \qquad a = 1.5, \quad b = 0.65, \quad g = h = 0.3. \tag{4}$$

The Lin panel of Figure 4 shows the resulting stability landscape. Of 8,100 grid points, 7,867 admitted a positive balanced equilibrium and 1,497 were locally stable. Thus stable positive balance occurred at 18.5% of all sampled points, or 19.0% of those admitting positive balance. The remaining 233 points lacked a positive synchronous equilibrium. These percentages describe this finite grid, not an intrinsic probability of stability for Lin's model.

Stable positive balance is therefore possible in Lin's model. For example, $(a, b, c, d, g, h) = (1, 2, 1, 1, 1, 1)$ gives $s_{\mathrm{L}} = 0.25$ and $\alpha \approx -0.095492$. Conclusions drawn from the all-ones baseline must not be extended to all parameter choices.

3.4 Status of motif analysis

The present results concern the full five-node system. A completed stability comparison of two-node motifs is not included here. Such a comparison must first specify whether the remaining nodes are deleted or clamped

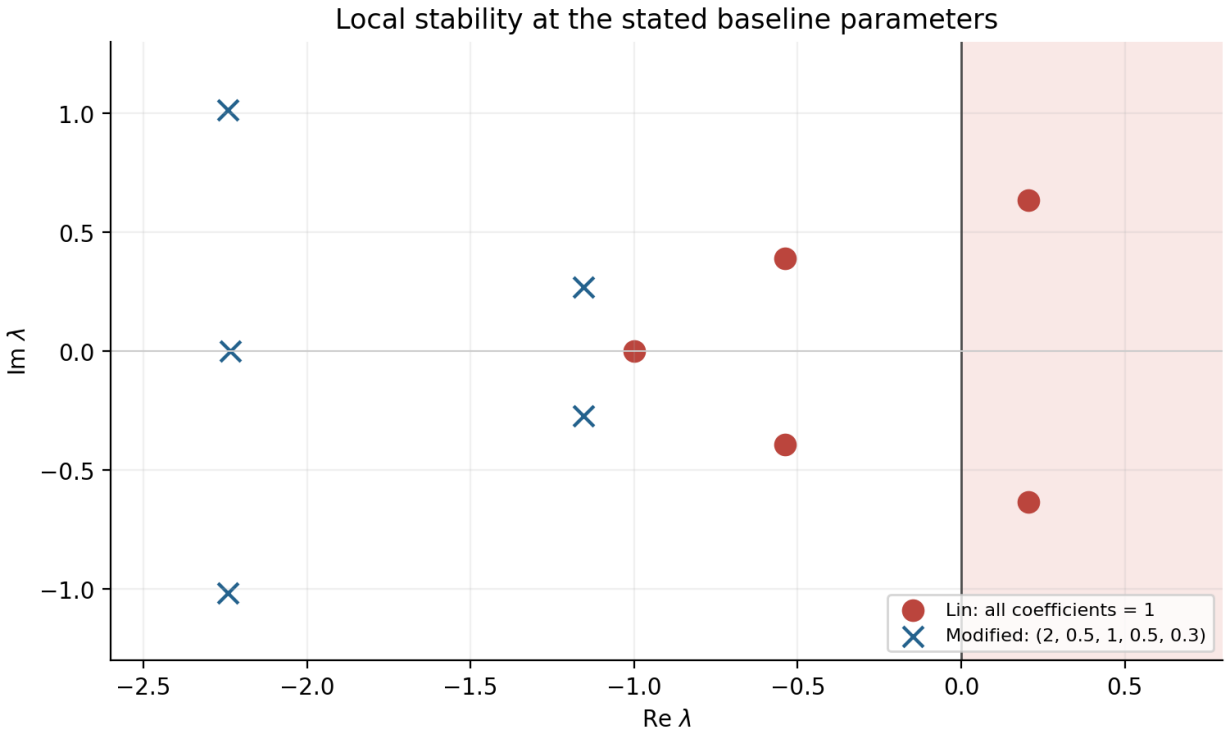


Figure 2: Jacobian eigenvalues at the stated positive balanced equilibria. Lin: $(a, b, c, d, g, h) = (1, 1, 1, 1, 1, 1)$ and $s_{\mathrm{L}} = 1/3$. Modified: $(a, k, c, g, h) = (2, 0.5, 1, 0.5, 0.3)$ and $s_{\mathrm{D}} = (3+\sqrt{59})/4$. The vertical line is $\mathrm{Re}\,\lambda = 0$. These are local stability results at different illustrative baselines, not a matched-parameter test of a single interaction mechanism.

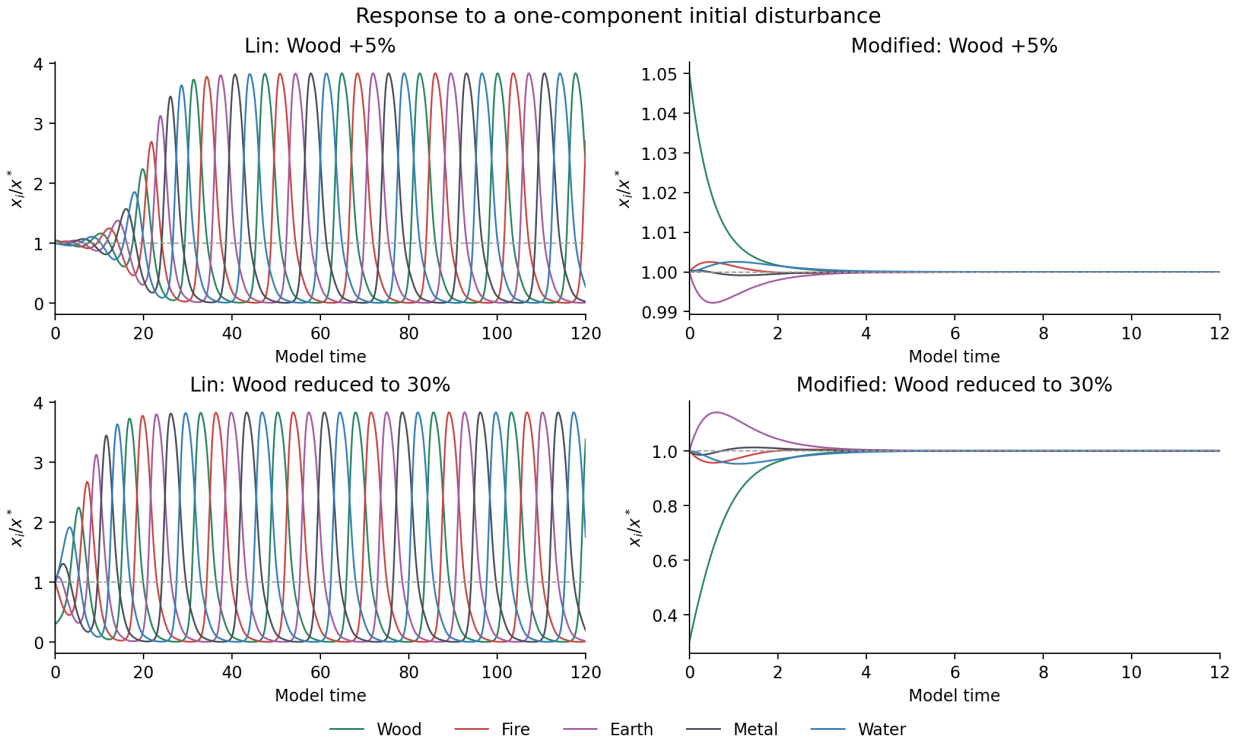


Figure 3: Responses to a one-component initial disturbance, using the baselines in Figure 2. Top: Wood increased by 5%; bottom: Wood reduced to 30% of its balanced value. Other components start at balance. Activities are divided by the relevant balanced level s. All runs extend to $t = 120$; the modified-model panels show only $0 \leq t \leq 12$ to make recovery visible. Each curve represents one of the five nodes. These are specified finite perturbations, not a basin-wide or globally uniform recovery test.

to fixed background activities, since these operations define different reduced systems. Motif dynamics are left for subsequent analysis.

4 A modified version of the model and its dynamics

4.1 Five-term model and nonnegative activities

The modified model considered in all results below is

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+2}} - hx_ix_{i-2}, \qquad i = 1, \ldots, 5. \tag{5}$$

We assume positive coefficients and nonnegative initial activities. The generating input is linear in predecessor activity and decreases with recipient activity. The target-dependent loss is moderated by target activity, while received restraint remains bilinear. At $x_i = 0$, the right-hand side is $cx_{i-1} \geq 0$, so the nonnegative orthant is invariant. Moreover, at a component attaining the maximum $X = \max_i x_i$, its derivative is bounded above by $aX - kX^2 + c$. Consequently, for $k > 0$, nonnegative solutions remain bounded by $\max\{\max_i x_i(0), B\}$, where

$$B = \frac{a + \sqrt{a^2 + 4kc}}{2k}. \tag{6}$$

This establishes a nonnegative, bounded activity domain; it does not imply global convergence to a particular equilibrium.

Equation (5) makes several simultaneous changes to Lin's model. It omits the direct child-loss term and changes both the predecessor input and the target-dependent loss. In particular, the partial derivative with respect to target activity is positive in the modified model, whereas it is negative in Lin's model. Also, the symbols g and h do not denote the same terms in the two equations: received restraint has coefficient d in Lin and h in the modified model. These differences must be considered when interpreting the plots.

4.2 Balanced equilibrium and baseline stability

The origin is an equilibrium. For a nonzero synchronous equilibrium $x = s\mathbf{1}$, the balanced level satisfies

$$(k+h)s^2 + (k+h-a)s - (a+c-g) = 0. \tag{7}$$

The maps use the larger positive root, when it exists. There is a unique positive root when $a + c > g$; other parameter choices can admit two positive roots, so this condition is sufficient but not necessary for positive balance. At a balanced level s, set

$$D_0 = a - 2ks - \frac{cs}{(1+s)^2} - \frac{g}{1+s} - hs.$$

The eigenvalue real parts are

$$\operatorname{Re}\lambda_j = D_0 + \frac{c}{1+s}\cos\theta_j + s\left(\frac{g}{(1+s)^2} - h\right)\cos(2\theta_j). \tag{8}$$

At $(a, k, c, g, h) = (2, 0.5, 1, 0.5, 0.3)$, $s_{\mathrm{D}} = (3 + \sqrt{59})/4 \approx 2.670286$ and $\alpha \approx -1.153724$. All five eigenvalues have negative real parts (Figure 2), establishing local exponential stability at this baseline.

For the same relative Wood perturbations used in Section 3, both modified-model trajectories returned to this equilibrium. Their final distances at $t = 120$ were approximately 3.3×10^{-12} and 1.2×10^{-11} (Figure 3). This supports recovery from the two tested initial conditions. Since the baselines have different coefficient values and equilibrium scales, the displayed decay and oscillation rates do not provide a calibrated comparison of physiological recovery times.

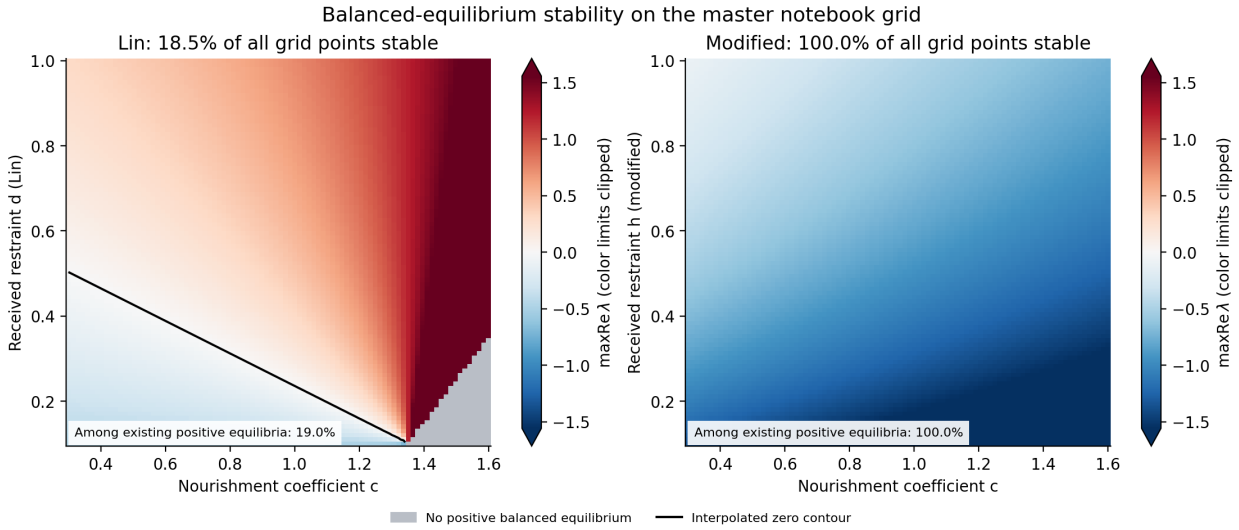


Figure 4: Stability maps on the verified 90×90 parameter grids. Horizontal axis: nourishment c. Vertical axis: received restraint, meaning d in Lin and h in the modified model. Lin uses $(a, b, c, d, g, h) = (1.5, 0.65, c, d, 0.3, 0.3)$; the modified model uses $(a, k, c, g, h) = (1.5, 0.65, c, 0.3, h)$. Colors show the largest real part of the Jacobian eigenvalues at the selected positive balanced equilibrium: blue indicates negative values and red positive values. Color limits are clipped, so the darkest colors also represent values beyond the displayed scale. Gray denotes no positive balanced equilibrium; the black line is an interpolated zero contour. Titles give the stable fraction among all grid points, while inset labels condition on existence of positive balance. These are specified slices of different complete models, including a fixed child-loss coefficient $g = 0.3$ in Lin; they do not isolate a single change in interaction law.

4.3 Stability maps and sensitivity to parameter sampling

For the modified-model panel of Figure 4, we used

$$c \in [0.3, 1.6], \qquad h \in [0.1, 1.0], \qquad a = 1.5, \quad k = 0.65, \quad g = 0.3, \tag{9}$$

on the same 90×90 grid. Every sampled point admitted a positive balanced equilibrium and was locally stable. Thus, in these specified slices, the modified model had a larger sampled region of stable positive balance than Lin's model. This does not establish stability for all parameters. For example, the earlier modified-model sweep with $a = 2$, $k = 0.5$, $g = 0.5$, $c \in [0.05, 2.5]$, and $h \in [0.05, 2.0]$ had 5,837 stable points out of 8,100 (72.1%); all those points admitted positive balance. The different percentages reflect different experiments.

We also summarized sensitivity to three illustrative parameter distributions (Figure 5 and Table 1). For a shared conceptual draw (S, A, C, R, T, L), representing self-growth, self-limitation, nourishment, received restraint, target-related loss, and child loss, the parameter assignments were

$$(a, b, c, d, g, h)_{\mathrm{L}} = (S, A, C, R, L, T), \qquad (a, k, c, g, h)_{\mathrm{D}} = (S, A, C, T, R).$$

The uniform scenario draws all six quantities independently from $[0.1, 2]$. The ordered scenario uses the same draws conditioned on $S > C > R > T$, without ordering A or L. The tiered scenario draws independently from $S \in [1, 2]$, $A \in [0.3, 1]$, $C \in [0.5, 1.5]$, $R \in [0.2, 0.8]$, and $T, L \in [0.1, 0.5]$. Each scenario uses 1,200 accepted draws and NumPy's default random-number generator with seed 3. The two models share each conceptual draw, but there is no child-loss coefficient in the modified model. This pairing specifies the experiment; it does not equate the strengths of different functional forms or remove the structural difference.

4.4 Interpretation and limitations of the preliminary comparison

The results establish a limited numerical observation: for the specified baselines, parameter slices, and sampling distributions, the modified model more often admits a locally stable positive balanced equilibrium,

Scenario	Model	Stable / eligible	Stable / eligible (%)	Stable / all (%)
Uniform	Lin	273 / 1195	22.8	22.8
	Modified	968 / 1029	94.1	80.7
Ordered	Lin	562 / 1192	47.1	46.8
	Modified	1103 / 1200	91.9	91.9
Tiered	Lin	267 / 1145	23.3	22.2
	Modified	1188 / 1200	99.0	99.0

Table 1: Stability frequencies under the stated sampling assumptions. "Eligible" means that the selected positive synchronous equilibrium exists; "all" means all 1,200 draws in that scenario. The counts retain both denominators, so failure to admit positive balance is not silently excluded. These are parameter-sampling results, not physiological probabilities or evidence of universal superiority.

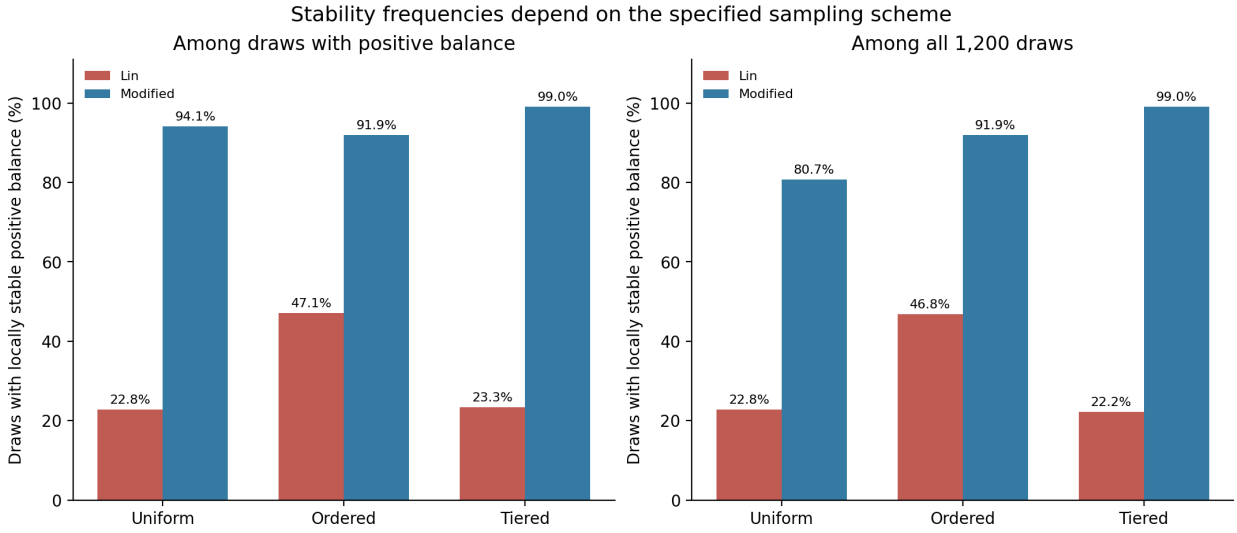


Figure 5: Stability frequencies under the uniform, ordered, and tiered sampling schemes described in the text, with 1,200 draws per scheme and seed 3. Left: the fraction with a locally stable positive balanced equilibrium among draws admitting the selected positive balance. Right: the fraction among all 1,200 draws. Exact counts are given in Table 1. These percentages depend on the sampling assumptions and eligibility rule; they are not probabilities that a model is correct or physiologically healthy. The comparison changes several mechanisms simultaneously and does not attribute the differences to one interaction law.

and it recovers from the two tested disturbances at its selected baseline. They do not show that demand modulation alone produces these differences. The comparison simultaneously changes interaction laws, removes the child-loss term, and, in the baseline examples, changes parameter values. Equal numerical coefficients multiplying different functions do not by themselves constitute a calibration. The size of a stable region also depends on the chosen coordinates, parameter window, and sampling measure.

A mechanism-focused comparison would first use a matched bilinear model with the same five terms as (5), and then vary the interaction laws separately. Removing the child-loss term from Lin's model gives this comparator after relabeling coefficients. Such a study would distinguish the effect of that removal from the effects of changing the input and target-dependent loss laws. It is not the experiment shown by the current Lin comparison diagrams. We therefore present these figures as preliminary descriptions of two complete models, without attributing their differences to a single cause or interpreting local stability as clinical validation. As in Section 3, the corresponding motif analysis remains to be completed.

5 Conclusion and Discussion

6 Acknowledgments

References

[1] H. Lin and J. Han. Analysis of Dynamic Five-Element Model. In *Proceedings of the 39th Chinese Control Conference*, pages 777–781, 2020. doi:10.23919/CCC50068.2020.9188438.